

# Out-of-hand nut consumption is associated with improved nutrient intake and health risk markers in US children and adults: National Health and Nutr...

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The purpose of this study was to determine the association of out-of-hand nut (OOHN) consumption with nutrient intake, diet quality, and the prevalence of risk factors for cardiovascular disease and metabolic syndrome. Data from 24-hour recalls from individuals aged 2+ years (n = 24,385) participating in the 1999-2004 National Health and Nutrition Examination Survey were used. The population was divided into children aged 2 to 11, 12 to 18, and adults 19+ years, and each group was dichotomized into OOHN consumers and nonconsumers. Out-of-hand nut consumers were defined as those individuals consuming  $\frac{1}{4}$  oz of nuts or more per d. Means, standard errors, and covariate-adjusted analyses of variance were determined using appropriate sample weights. Diet quality was determined using the Healthy Eating Index-2005. Significance was set at  $P < .05$ . The percent of OOHN consumers increased with age:  $2.1\% \pm 0.3\%$ ,  $2.6\% \pm 0.3\%$ ,  $6.5\% \pm 0.5\%$ , and  $9.6\% \pm 0.5\%$  those aged 2 to 11, 12 to 18, 19 to 50, and 51+ years, respectively. The 2 latter groups were combined into a single group of consumers aged 19+ years for subsequent analyses. Consumers of OOHN from all age groups had higher intakes of energy, monounsaturated and polyunsaturated fatty acids, dietary fiber, copper, and magnesium and lower intakes of carbohydrates, cholesterol, and sodium than did nonconsumers. Diet quality was higher in OOHN consumers of all age groups. In children aged 2 to 11 years, consumers had a higher prevalence of overweight/obesity. In those aged 12 to 18 years, weight and percent overweight were lower in consumers. Adult consumers had higher high-density lipoprotein cholesterol, red blood cell folate, and serum folate levels and lower insulin, glycohemoglobin, and C-reactive protein levels than did nonconsumers. Adult consumers also had a 19% decreased risk of hypertension and a 21% decreased risk of low high-density lipoprotein cholesterol levels. Data suggested that OOHN consumption was associated with improved nutrient intake, diet quality, and, in adults, a lower prevalence of 2 risk factors for metabolic syndrome. Consumption of OOHN, as part of a healthy diet, should be encouraged by health professionals.

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